2020A F=ma Exam: Problem 12

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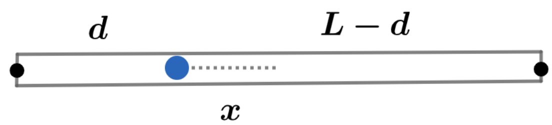

Recall the angular frequency of a physical pendulum is given by

$$\omega = \sqrt{\frac{mgd}{I}}$$

where m is the total mass of the object, d is the distance from the pivot to the center of mass, and I is the moment of inertia about the pivot. In our case,

$$I = md^2$$
$$\omega = \sqrt{\frac{g}{d}}$$

By definition,

$$T = \frac{2\pi}{\omega} \propto \sqrt{d}$$

so we have

$$\frac{T}{\sqrt{d}} = \frac{2T}{\sqrt{L-d}}$$
$$L = 5d$$

Thus,

$$x = \frac{L}{2} - d = L\left(\frac{1}{2} - \frac{1}{5}\right) = \frac{3L}{10}$$

so the answer is $\boxed{\text{D}}$.